

Deep Unfolding Network for Efficient Mixed Video Noise Removal

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Abstract—Existing image and video denoising algorithms have focused on removing homogeneous Gaussian noise. However, this assumption with noise modeling is often too simplistic for the characteristics of real-world noise. Moreover, the design of network architectures in most deep learning-based video denoising methods is heuristic, ignoring valuable domain knowledge. In this paper, we propose a model-guided deep unfolding network for the more challenging and realistic mixed noise video denoising problem, named DU-MVDnet. First, we develop a novel observation model/likelihood function based on the correlations among adjacent degraded frames. In the framework of Bayesian deep learning, we introduce a deep image denoiser prior and obtain an iterative optimization algorithm based on the maximum a posteriori (MAP) estimation. To facilitate end-to-end optimization, the iterative algorithm is transformed into a deep convolutional neural network (DCNN)-based implementation. Furthermore, recognizing the limitations of traditional motion estimation and compensation methods, we propose an efficient multistage recursive fusion strategy to exploit temporal dependencies. Specifically, we divide video frames into several overlapping groups and progressively integrate these frames into one frame. Toward this objective, we implement a multiframe adaptive aggregation operation to integrate feature maps of intra-group with those of intergroup frames. Extensive experimental results on different video test datasets have demonstrated that the proposed model-guided deep network outperforms current state-of-the-art video denoising algorithms such as FastDVDnet and MAP-VDNet.

Index Terms—Mixed video noise removal, model-guided design, deep convolutional neural network, multistage progressive fusion.

I. INTRODUCTION

THE denoising problem is one of the most extensively studied inverse problems in image and video processing. In the open literature, most image and video denois-

ing methods aim to eliminate homogeneous Gaussian noise [1], [2], [3], [4], [5], [6], [7], [8], [9], [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], [25], [26], [27], [28], [29], [30], [31], [32], [33], [34], [35], [36], [37], [38], [39], [40], [41], which is simple and convenient. However, the practical distribution of real-world noise is often heterogeneous and more complicated than what an oversimplified Gaussian noise model can characterize. Moreover, images and videos could be contaminated by several types of noise during acquisition and transmission processes (e.g., the mixture of additive white Gaussian noise (AWGN) and impulse noise [42]). Specifically, AWGN could be caused by imperfections in sensor readout circuitry and impulse noise could be introduced by abrupt disturbances or shot noise [43]. Unlike Gaussian denoising, mixed noise removal is a more difficult and under-researched problem in the literature. Due to the heterogeneous nature of mixed noise, it is hard to distinguish different noise types and adapt the model spatially from pixel to pixel.

Existing mixed noise removal methods [44], [45], [46], [47], [48] generally follow two different lines of thought. The first class of methods [49], [50], [51], [52] employ a two-stage strategy that first detects corrupted pixels and recovers the clean image by median filtering. However, the accuracy of detection would be reduced especially when the intensity of noise is high, which would destroy the denoising performance subsequently. The second class of methods uses the robust ℓ_1 -norm [42], [53], [54] to penalize the fidelity terms of impulse noise and iteratively remove mixed noise. Nevertheless, there are still several problems that need to be addressed for the mixed noise removal task. First, since the most popular methods [42], [44], [45], [46], [47], [48], [49], [50], [51], [52], [53], [54], [55], [56], [57], [58] for mixed noise removal are model-based methods and hyperparameters are manually adjusted, denoising performance is somewhat limited. Second, several deep learning-based methods [59], [60] have been developed to remove mixed noise, but these methods are completely heuristic and ignore significant domain knowledge. Third, the mixed noise removal techniques for nature images have been studied for decades, while the mixed video noise removal problem is still under-researched; how to integrate temporal correlations is an essential problem in video processing.

In this paper, we propose a model-guided deep unfolding network for mixed noise video denoising, named DU-MVDnet, with both good generalization and interpretation properties. First, we develop a robust observation

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model/likelihood function for the problem of mixed video noise removal. To facilitate the end-to-end optimization, we introduce a deep denoiser prior and unfold the iterative processes into a deep network. Instead of using expensive frame-by-frame alignment, we propose an efficient multistage progressive fusion strategy. To achieve adaptation along the temporal axis, we classify input noisy frames into several groups and integrate intra-group frames and inter-group frames for multistage fusion, respectively. During the integration of features from intra-group and inter-group frames, we exploit DCNNs to estimate pixel-wise filters, and adaptively aggregate multi-frame features through spatiotemporal weighted averaging. Extensive experimental results for mixed video noise removal on two test datasets have verified the effectiveness of the proposed model-guided method, and extended experiments for video decaptioning demonstrate the generalization and robustness of DU-MVDnet. To the best of our knowledge, this is the first deep learning-based method to solve the problem of mixed video noise removal. Compared to Gaussian denoising methods [23], [24], [26], [27], [30], [61], [62], the proposed method can deal with more realistic and complicated noisy scenarios. On the other hand, the proposed method could greatly encourage the performance benefits of aggregation of temporal information compared to existing mixed noise removal methods [42], [47], [54], [59]. The main contributions of this paper could be summarized as follows.

- Effective deep model-guided network. To solve the mixed video noise removal problem, we propose a robust iterative optimization algorithm based on the MAP estimation. Guided by principled solutions, we design a deep unfolding network where each operation could be effectively implemented.
- Multistage progressive fusion strategy. To reduce computational costs and improve denoising performance, we develop a multistage progressive fusion strategy. In particular, adaptive temporal aggregation is realized through weighting filters estimated by DCNNs.
- Superior denoising performance. The proposed method outperforms state-of-the-art video denoising methods, and its robustness has been proved through experiments for video decaptioning.

II. RELATED WORKS

A. AWGN Removal

The Gaussian denoising problem has been widely studied in recent decades. Among these algorithms, traditional model-based methods [1], [2], [3], [4], [5], [6], [7], [8], [9], [10], [11], [12], [13], [23], [24], [25], [26], [27], [28], [29], [30], [31], [32], [33], [34] are one kind of classic method. From the view of Bayesian optimization, model-based denoising methods develop effective objective functions by constructing Gaussian likelihood terms and various regularization terms to obtain corresponding iterative solutions. Among model-based image denoising algorithms, nonlocal means [2], [3] and sparse representation [6], [8] are the most widely used image priors. In particular, nonlocal methods exploit the self-similarity property in natural images, meanwhile, sparse

representation-based methods focus on constructing sparse coefficients to recover the clean image. Recently, many deep learning-based Gaussian denoising methods [14], [15], [16], [17], [18], [19], [20], [21], [22], [35], [36], [37], [38], [39], [40], [41], [63], [64], [65], [66], [67] have been proposed with regard to DCNN as a black box. To benefit from extensive external datasets and strong nonlinear fitting capabilities of DCNNs, these methods have obtained great performance advances and have gradually dominated this field. To improve the performance of denoising and the stability of training for DCNN-based methods, many effective techniques have been introduced, such as residual learning [14], memory network [15], and robust noise map [16]. Furthermore, several model-guided methods [18], [68] have been developed to integrate domain knowledge.

B. Impulse Noise Removal

The most classic method of removing impulse noise is medium filtering [69], this algorithm could effectively remove IN by replacing corrupted pixels with the medium value in a local window. However, the medium filtering method would destroy the details of the image and lead to blurring, especially when the IN level is high. To address this problem, many improved algorithms [70], [71], [72], [73], [74], [75], [76], [77], [78], [79], [80], [81] have been proposed. A category of methods first detects the locations of damaged pixels and then recovers these pixels, such as the switching median filter [73], the adaptive median filter (AMF) [74], the tristate median filter [75], the adaptive center weighted median filter (ACWMF) [76], the conditional signal adaptive median filter [77] and the directional weighted median filter [78]. Rather than collecting the locations of corrupted pixels, several methods smooth image details, including the weighted median filter [70], the center-weighted median filter [72] and the multi-state median filter [71]. Besides, considering the sparse properties of IN, some methods [82], [83] integrate the ℓ_1 -norm data fidelity term with IN and recover the clean image through iterative algorithms.

C. Mixed Noise Removal

Regarding the sophisticated mixed noise removal problem, the critical issue is to distinguish different noise categories and determine the noise intensities of image pixels. Since different kinds of noise follow various distribution characteristics, previous denoising works usually exploit the ℓ_2 norm fidelity term for AWGN [2], [6], [8], [27], [33] and the ℓ_1 norm fidelity term for IN [82], [84]. A two-stage mixed noise removal method was first proposed in [55], and a more efficient version was developed in [56]. The authors of [49] proposed a novel approach to minimization of $\ell_1 - \ell_0$. To remove mixed noise, ℓ_1 and ℓ_2 fidelity terms together with TV regularization were proposed in [51]. Furthermore, a kernel estimation approach [57] associated with classification was developed to remove mixed noise. An adaptive denoising method was proposed in [58] where noise parameter estimation and noise classification could be implemented effectively. In [52], dictionary learning, sparse representations, and noise parameter

estimation were integrated into a weighted dictionary learning method to recover the clean image. In [54], a Laplacian-scale mixture model was proposed together with nonlocal low-rank regularization. Recently, a CNN regularizer was introduced in [85], associated with a variational denoising method. To benefit from deep learning techniques, [59] proposed a generative adversarial network to denoise the mixed noise, and a DCNN based on non-local means [60] was developed to remove the mixture of AWGN and IN.

D. Video Denoising

The most essential problem for video denoising is designing efficient alignment and fusion modules to take advantage of neighboring frames and improve denoising performance. Since the generation of Gaussian noise is convenient, most video denoising methods [23], [24], [25], [26], [27], [28], [29], [30], [31], [32], [33], [34], [35], [36], [37], [38], [39], [40], [41], [64], [65], [66], [67] focus on removing the Gaussian noise.

Similarly to image denoising methods, video denoising methods could be roughly classified into model-based methods [23], [24], [25], [26], [27], [28], [29], [30], [31], [32], [33], [34] and deep learning-based methods [35], [36], [37], [38], [39], [40], [41], [64], [65], [66], [67]. On the one hand, most model-based video denoising algorithms [27], [28], [30] are extended from classic image denoisers by exploiting temporal dependencies, for example, the nonlocal image denoising method [2] expands to [86], and the classic V-BM4D [30] method is extended from BM3D [3]. On the other hand, DCNN-based video denoising methods [35], [36], [38], [40], [41], [65], [66], [67], [87], [88], [89] have demonstrated superior denoising performance. To deal with large motion scenarios, some methods [35], [41], [65], [67] employ explicit motion estimation and compensation operations. For example, [35] has proposed a task-oriented optical flow estimation method using a multi-scale DCNN structure. Several video denoising algorithms [41], [87] have been developed to benefit from the recurrent architecture, and alignment operations are mainly conducted between two adjacent frames. To increase the interpretability of deep networks, [67] has proposed a MAP-based video denoising method, which has obtained excellent denoising performance. Furthermore, some denoising algorithms opt to align adjacent frames in an implicit manner. [88] predicts spatially varying kernels to perform temporal aggregation and denoise operations. To reduce computing complexity, some denoising methods [36], [38], [40], [66] directly handle noisy frames, leaving out the time-consuming alignment operations. In [40], authors have developed an efficient multi-stage fusion strategy to progressively restore the clean frame. To avoid complicated alignment operations, [66] obtains a real-time network implementation by only exploiting the temporal information from the previous frame.

III. PROPOSED METHOD

In this section, we first introduce an effective iterative optimization algorithm for mixed video noise removal based on the MAP estimation framework. Instead of expensive motion

estimation and compensation, we develop an efficient multi-stage progressive fusion strategy for the restoration of video frames. Then we unfold the process of iterative optimization into a deep network implementation to facilitate end-to-end training.

A. MAP Estimation for Mixed Video Noise Removal

The mixed noise removal task aims to recover clean video frames $\mathbf{x}_j \in \mathbb{R}^N$ from noisy observations $\mathbf{y}_j \in \mathbb{R}^N$, following the setting in [54], the degraded model of a single video frame can be described as

$$\mathbf{y}_j = \mathbf{x}_j + \mathbf{s}_j + \mathbf{n}_j, \quad (1)$$

where $\mathbf{s}_j \in \mathbb{R}^N$ denotes outliers caused by impulse noise, and $\mathbf{n}_j \in \mathbb{R}^N$ denotes the AWGN of the j -th video frame, $j = -J, \dots, J$. To integrate temporal dependency, we introduce a warping matrix $\mathbf{W}_j \in \mathbb{R}^{N \times N}$ [67], [90] to map neighboring frames $\mathbf{x}_j$ to the reference frame $\mathbf{x}$, and the observation model of noisy frames can be expressed accordingly as

$$\mathbf{x} = \mathbf{W}_j \mathbf{x}_j = \mathbf{W}_j \mathbf{y}_j + \tilde{\mathbf{s}}_j + \tilde{\mathbf{n}}_j, \quad (2)$$

where $\tilde{\mathbf{s}}_j = -\mathbf{W}_j \mathbf{s}_j$ and $\tilde{\mathbf{n}}_j = -\mathbf{W}_j \mathbf{n}_j$. Note that the warping matrix $\mathbf{W}_j$ can be interpreted as a global approximation of standard image registration techniques, we can develop a joint MAP estimation to simultaneously estimate the reference frame $\mathbf{x}$ and the outliers $\tilde{\mathbf{s}}_j$ of impulse noise,

$$\begin{aligned} (\mathbf{x}, \tilde{\mathbf{s}}_j) &= \underset{\mathbf{x}, \tilde{\mathbf{s}}_j}{\operatorname{argmax}} \sum_{j=-J}^J \log P(\mathbf{x}, \tilde{\mathbf{s}}_j | \mathbf{y}_j), \\ &= \underset{\mathbf{x}, \tilde{\mathbf{s}}_j}{\operatorname{argmax}} \sum_{j=-J}^J \log P(\mathbf{y}_j | \mathbf{x}, \tilde{\mathbf{s}}_j) + \log P(\mathbf{x}) \\ &\quad + \sum_{j=-J}^J \log P(\tilde{\mathbf{s}}_j). \end{aligned} \quad (3)$$

Based on the proposed observation model in Eq. (2), we can rewrite the Gaussian likelihood function as

$$P(\mathbf{y}_j | \mathbf{x}, \tilde{\mathbf{s}}_j) \propto \exp\left(-\frac{1}{2\sigma_n^2} \|\mathbf{x} - \mathbf{W}_j \mathbf{y}_j - \tilde{\mathbf{s}}_j\|_2^2\right), \quad (4)$$

where σ_n is the standard deviation of AWGN. With the assumption about the sparse characteristics of impulse noise [53], [54], [82], we consider a ℓ_1 -norm regularization to model the prior term of $\tilde{\mathbf{s}}_j$ as follows (its effectiveness will be verified in Sec. IV-E)

$$P(\tilde{\mathbf{s}}_j) \propto \exp\left(-\frac{1}{\theta} \|\tilde{\mathbf{s}}_j\|_1\right). \quad (5)$$

Instead of analytical modeling for the prior term $\log P(\mathbf{x})$, we advocate a DCNN-based denoiser prior $f(\cdot)$ [18], [67] to leverage the latest advances in deep learning-based image/video restoration,

$$P(\mathbf{x}) \propto \exp(-\lambda \|\mathbf{x} - \mathbf{v}\|_2^2), \quad \text{where } \mathbf{v} = f(\mathbf{x}). \quad (6)$$

By substituting the Gaussian likelihood function of Eq. (4) and the prior terms of Eqs. (5) and (6) in the MAP estimation

of Eq. (3), the overall objective function can be described as follows,

$$(\mathbf{x}, \tilde{\mathbf{s}}_j) = \underset{\mathbf{x}, \tilde{\mathbf{s}}_j}{\operatorname{argmin}} \sum_{j=-J}^J \eta \|\mathbf{x} - \mathbf{W}_j \mathbf{y}_j - \tilde{\mathbf{s}}_j\|_2^2 + \lambda \|\mathbf{x} - \mathbf{v}\|_2^2 + \sum_{j=-J}^J \delta \|\tilde{\mathbf{s}}_j\|_1, \text{ where } \mathbf{v} = f(\mathbf{x}), \quad (7)$$

where hyperparameters $\eta = \frac{1}{2\sigma_n^2}$ and $\delta = \frac{1}{\theta}$. Based on the principle of iterative optimization, the mixed video noise removal problem could be decomposed into two subproblems as

$$\tilde{\mathbf{s}}_j = \underset{\tilde{\mathbf{s}}_j}{\operatorname{argmin}} \sum_{j=-J}^J \eta \|\mathbf{x} - \mathbf{W}_j \mathbf{y}_j - \tilde{\mathbf{s}}_j\|_2^2 + \sum_{j=-J}^J \delta \|\tilde{\mathbf{s}}_j\|_1, \quad (8a)$$

$$\mathbf{x} = \underset{\mathbf{x}}{\operatorname{argmin}} \sum_{j=-J}^J \eta \|\mathbf{x} - \mathbf{W}_j \mathbf{y}_j - \tilde{\mathbf{s}}_j\|_2^2 + \lambda \|\mathbf{x} - \mathbf{v}\|_2^2, \quad (8b)$$

where $\mathbf{v} = f(\mathbf{x})$.

According to Eq. (8), the $\tilde{\mathbf{s}}_j$ -subproblem could be solved by a soft-thresholding function, and the $\mathbf{x}$ -subproblem is a quadratic optimization and can be solved in closed-form, the corresponding iterative solution can be written as

$$\tilde{\mathbf{s}}^{(t+1)} = \mathbf{S}_\tau(\mathbf{x}^{(t)} - \bar{\mathbf{y}}' + \epsilon), \quad (9a)$$

$$\begin{aligned} \mathbf{x}^{(t+1)} &= \frac{\eta}{N} \sum_{j=-J}^J (\mathbf{W}_j \mathbf{y}_j + \tilde{\mathbf{s}}_j) + \frac{\lambda}{N} f(\mathbf{x}^{(t)}), \\ &= (1 - \alpha)(\bar{\mathbf{y}}' + \tilde{\mathbf{s}}^{(t+1)}) + \alpha f(\mathbf{x}^{(t)}), \end{aligned} \quad (9b)$$

where $\mathbf{S}_\tau(\cdot)$ is a soft-thresholding operator with threshold τ , $\bar{\mathbf{s}}$ denotes the average outliers of all neighboring frames, i.e., $\bar{\mathbf{s}} = \frac{1}{2J+1} \sum_{j=-J}^J \tilde{\mathbf{s}}_j$, and $\bar{\mathbf{y}}'$ denotes the average of aligned neighboring frames, i.e., $\bar{\mathbf{y}}' = \frac{1}{2J+1} \sum_{j=-J}^J \mathbf{W}_j \mathbf{y}_j$, $N = \eta(2J+1) + \lambda$, the weighting coefficient $\alpha = \frac{\lambda}{\eta(2J+1) + \lambda}$, $f(\cdot)$ is a DCNN-based image denoiser, and the constant ϵ is for numerical stability. According to the solution in Eq. (9), the mixed video noise removal problem could be effectively solved by iteratively computing $\bar{\mathbf{s}}$ and $\mathbf{x}$. In particular, we use the aggregate frame $\bar{\mathbf{y}}'$ to initialize $\mathbf{x}^{(0)}$, i.e., $\mathbf{x}^{(0)} = \bar{\mathbf{y}}'$, the initial $\mathbf{x}^{(0)}$ would be fed to soft-thresholding and denoising modules to compute $\tilde{\mathbf{s}}^{(1)}$ and $f(\mathbf{x}^{(0)})$, respectively. In our implementation, we use a DCNN to learn a pixel-wise threshold map τ for each iteration to obtain a more accurate calculation of $\tilde{\mathbf{s}}^{(t)}$.

Regarding the prior term $\log P(\mathbf{x})$, instead of popular low-rank/sparse priors [6], [8], [91], we employ a DCNN-based denoiser prior for the following reasons. First, the denoising methods that employ low-rank/sparse priors [6], [8], [91] are generally model-based methods, these methods usually need to manually adjust appropriate hyper-parameters to obtain competitive denoising performance, which is laborious. By applying a deep learning-based denoiser prior, the proposed method could deploy iterative optimization into a deep network, and all hyper-parameters could be adaptively updated throughout

Algorithm 1 Proposed MAP Estimation for Mixed Video Noise Removal

• **Initialization:**

- (1) Set α , $t = 0$;
- (2) Align $\mathbf{y}_j$ to $\mathbf{y}$ using $\mathbf{W}_j$, and estimate the aggregate frame as $\bar{\mathbf{y}}' = \frac{1}{2N+1} \sum_{j=-J}^J \mathbf{W}_j \mathbf{y}_j$;
- (3) Initialize $\mathbf{x}$ as $\mathbf{x}^{(0)} = \bar{\mathbf{y}}'$;
- **While** does not converge **do**
- (4) Compute denoise prior $f(\mathbf{x}^{(t)})$;
- (5) Learn a pixel-wise threshold map τ ;
- (6) Estimate outliers $\tilde{\mathbf{s}}^{(t+1)} = \mathbf{S}_\tau(\mathbf{x}^{(t)} - \bar{\mathbf{y}}' + \epsilon)$;
- (7) Compute $\mathbf{x}^{(t+1)} = (1 - \alpha)(\bar{\mathbf{y}}' + \tilde{\mathbf{s}}^{(t+1)}) + \alpha f(\mathbf{x}^{(t)})$;
- (8) $t = t + 1$.

End while

Output: $\mathbf{x}^{(T)}$

the training. Furthermore, model-based methods [6], [8], [91] require multiple iterations to recover clean frames. Unlike model-based methods, the proposed unfolding network could greatly reduce the number of iterations, benefiting from the introduction of deep denoiser prior and end-to-end optimization. Lastly, the iterative solutions of model-based methods are completely dependent on domain knowledge, while the proposed method could make full use of extensive external datasets using a deep learning technique to further prompt denoising performance.

The proposed iterative algorithm for mixed video noise removal has been summarized in Algorithm 1. To facilitate end-to-end optimization, we use DCNNs to calculate the aggregate frame $\bar{\mathbf{y}}'$ and learn the threshold map τ and the denoiser $f(\cdot)$. In the next section, we introduce an efficient multistage progressive fusion strategy.

B. Efficient Multistage Progressive Fusion Strategy

In recent years, optical flow estimation [92], [93] and deformable convolutional network (DCN) [94], [95], [96] have been widely used for motion estimation and compensation in video processing [35], [67], [96]. However, these popular techniques have several shortcomings. For example, optical flow estimation methods usually employ an expensive multiscale network architecture, which is time-consuming. Regarding deformable convolution, the critical motion compensation process is completed through the offsets learned by the convolutional layers. However, the training process of the deformable convolutional network is not always stable, and the computational complexity is comparatively high due to the additional convolutional layers to compute the offsets.

In contrast, the effectiveness of multistage architectures in FastDVDnet [40] and TGA [97] has been shown to implicitly handle motions with different rates. Inspired by the two-step denoising and fusion frameworks, we propose an efficient multistage progressive fusion strategy to avoid explicit motion estimation and compensation. Different from FastDVDnet and TGA methods, we use DCNNs to estimate pixel-wise filters and adaptively aggregate feature maps. The detailed structure is demonstrated in Fig. 2, and it should be noted that the

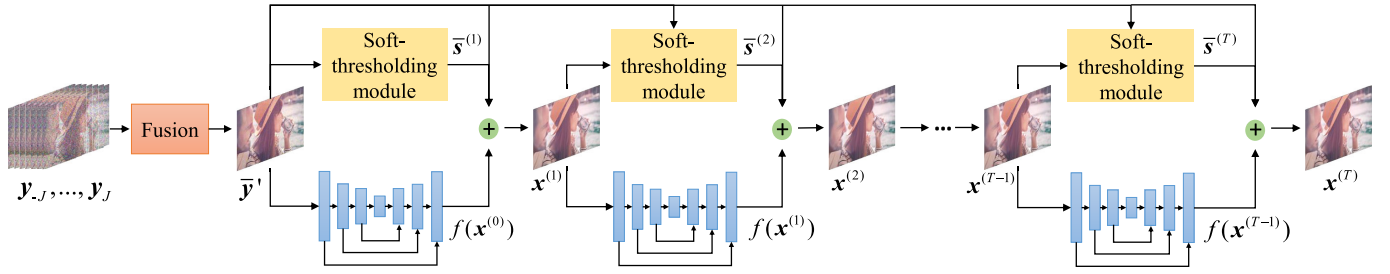


Fig. 1. The overall architecture of the proposed deep unfolding network.

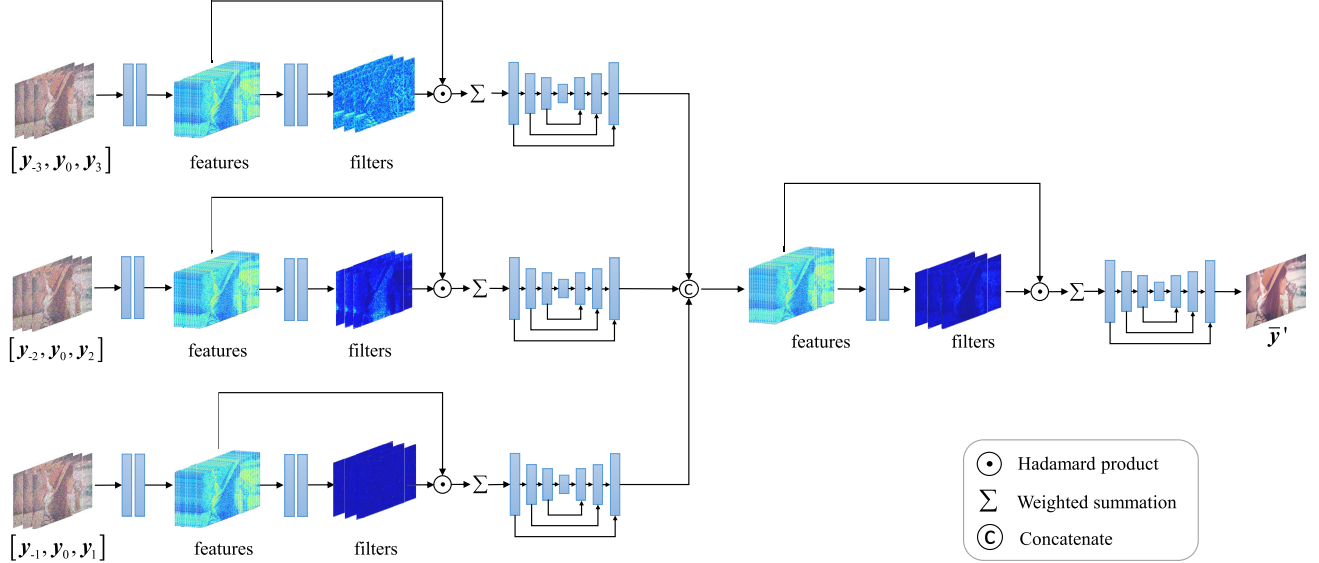


Fig. 2. The architecture of the fusion module.

target of the developed fusion module is to estimate $\bar{y}'$ in Eqs. (9a) and (9b). As shown in Fig. 2, to deal with videos with various frame rates, we first classify input noisy frames into several groups with different frame intervals. For each branch, we first compute filters to adaptively aggregate feature maps of adjacent frames, and then we exploit Unet to further denoise integrated feature maps. Subsequently, the feature maps between the groups are aggregated by concatenation operation, and a similar adaptive fusion is performed to obtain the final output $\bar{y}'$. Through the prediction of pixel-wise filters, the adaptive temporal aggregation of intragroup and intergroup features could be efficiently implemented. Compared to existing video denoising methods, the proposed method not only reduces the computational burdens but also boosts denoising performance through the effective incorporation of temporal information.

C. Deep Unfolding Network Implementation

In this section, we develop a deep unfolding network guided by Algorithm 1. The general framework is demonstrated in Fig. 1, it could be observed that the neighboring frames $y_{-J}, \dots, y_J$ are first integrated into one frame $\bar{y}'$ through the fusion module, and the aggregate frame is used as the initialization of $x^{(t)}$. For each iteration, $x^{(t)}$ is fed into two parallel branches; one is the soft-thresholding module that aims at computing $\bar{s}^{(t+1)}$, and the other branch is an Unet denoiser to estimate $f(x^{(t)})$. Finally, $x^{(t+1)}$ could be updated

by the weighted average operation, as shown in Eq. (9b). Next, we will provide a detailed introduction of each module.

Multi-stage progressive fusion module: This module performs step (2) in Algorithm 1, that is, aligning and integrating several neighboring frames and obtaining the aggregate frame. Inspired by group-based fusion methods [40], [97], we propose an efficient multistage progressive fusion strategy to integrate adjacent frames while avoiding time-consuming motion estimation and motion compensation. The specific structure of the fusion module is presented in Fig. 2, and we set $J = 3$ for clear visualizations, which is consistent with our implementation. As demonstrated in Fig. 2, we first divide 7 noisy frames into 3 groups, and each group includes the reference frame. In particular, we collect neighboring frames that have the same temporal distances from the reference frame into a group. In principle, the developed architecture can handle different rates of motion. Specifically, regarding intragroup integration, we first concatenate noisy frames to extract features and send these features into 2 convolutional layers to predict pixel-wise filters and achieve adaptive multiframe aggregation. After that, a weighting fusion operation is implemented in the feature domain through dot product and summation operations between estimated filters and extracted features; then the integrated features are sent to Unet [98] for further refinement. After the first stage of intragroup fusion, a similar intergroup fusion process is implemented in the second stage, and the target frame $\bar{y}'$ could be recovered using the aggregate features of the first stage.

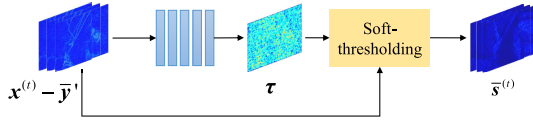


Fig. 3. The architecture of the pixel-wise soft thresholding module.

Denoising module: In this work, we use a DCNN-based denoiser to benefit from the deep learning technique, this module conducts step (4) in Algorithm 1. Similarly to [18] and [67], we use an Unet denoiser to compute $f(\mathbf{x}^{(t)})$ in Eq. (9b), and the specific structure of Unet is the same as that of [67].

Soft-thresholding module: This component implements steps (5) and (6) in Algorithm 1, and the structure of the soft-thresholding module is described in Fig. 3. Specifically, we first feed the difference value of the current estimation $\mathbf{x}^{(t)}$ and the integrated frames $\bar{\mathbf{y}}'$, i.e., $\mathbf{x}^{(t)} - \bar{\mathbf{y}}'$, into a DCNN to compute the pixel-wise thresholding map τ , then perform the soft thresholding computation in Eq. (9a) to obtain the updated $\bar{\mathbf{s}}^{(t+1)}$. In particular, the pixel-wise soft-thresholding computation could be written as

$$\bar{s}_p^{(t+1)} = \begin{cases} x_p^{(t)} - \bar{y}_p' + \tau_p, & x_p^{(t)} - \bar{y}_p' < -\tau_p \\ 0, & -\tau_p \leq x_p^{(t)} - \bar{y}_p' \leq \tau_p \\ x_p^{(t)} - \bar{y}_p' - \tau_p, & x_p^{(t)} - \bar{y}_p' > \tau_p \end{cases} \quad (10)$$

where p denotes the index of pixel locations. Based on the previous iterative output $\mathbf{x}^{(t)}$, the average of aligned adjacent frames $\bar{\mathbf{y}}'$ and the thresholding map τ predicted by a DCNN, the pixel-wise soft thresholding computation of Eq. (10) could be effectively implemented. In practice, we first initialize the average outliers $\bar{\mathbf{s}}^{(t+1)}$ with a zero matrix, then search for image pixels satisfying $x_p^{(t)} - \bar{y}_p' < -\tau_p$ and $x_p^{(t)} - \bar{y}_p' > \tau_p$, and the soft-thresholding computation could be effectively conducted by assigning $x_p^{(t)} - \bar{y}_p' + \tau_p$ and $x_p^{(t)} - \bar{y}_p' - \tau_p$ to the collected pixels, respectively. Note that there are two special points that are not differentiable, i.e., $x_p^{(t)} - \bar{y}_p' = -\tau_p$ and $x_p^{(t)} - \bar{y}_p' = \tau_p$. In our implementation, we explicitly set the derivatives of two non-differentiable points to 0. Regarding the structure of DCNN predicting the thresholding map τ , the DCNN consists of 2 convolutional layers and 5 residual blocks, 2 convolutional layers with 3×3 kernels are distributed at the head and tail of the DCNN respectively to alter the number of channels, and each residual block contains 2 convolutional layers with ReLU nonlinearity, each convolutional layer generates 32-channel feature maps with 3×3 kernels.

As demonstrated in Fig. 1, $\mathbf{x}^{(t+1)}$ could be updated at the end of each iteration by implementing the weighting fusion operation in Eq. (9b). Generally speaking, the overall network in Fig. 1 implements T iterations, $\bar{\mathbf{s}}^{(t+1)}$ and $\mathbf{x}^{(t+1)}$ could be effectively computed in each iteration. Specifically, the fusion module, the soft-thresholding module and the denoise module could be efficiently implemented by DCNNs. Besides, the hyper-parameter in Eq. (9b) is first initialized with reasonable decimal numbers, e.g., $\alpha = 0.5$, and then updated along the network training. Therefore, the overall network could be optimized through end-to-end training.

Differences with the MAP-VDNet method [67]: To incorporate domain knowledge, the proposed network possesses

a deep unfolding architecture similar to the MAP-VDNet method [67], but our approach is essentially different from MAP-VDNet in the following aspects. First, since the proposed method focuses on removing the more realistic mixed noise, we introduce several variables, e.g., s_j and n_j , to represent various characteristics of different types of noise in the degraded model of Eq. (1). Furthermore, the proposed method has developed a multi-stage progressive fusion strategy to effectively integrate temporal information, while the MAP-VDNet method [67] needs to implement time-consuming motion estimation and compensation processes. Finally, by accurately modeling the characteristics of different types of noise, the proposed DU-MVDnet has obtained great performance improvements with fewer computational burdens over MAP-VDNet when dealing with the mixed video noise removal problem.

D. Network Training

The overall deep unfolding network is trained by ℓ_1 loss function as follows,

$$\Theta = \underset{\Theta}{\operatorname{argmin}} \sum_{j=-J}^J \|\mathcal{F}(\mathbf{y}_j; \Theta) - \mathbf{x}\|_1 \quad (11)$$

where $\mathbf{x}$ denotes the ground truth of the reference frame, $\mathbf{y}_j (j = -J, \dots, J)$ are neighboring noisy frames, and $\mathcal{F}(\cdot; \Theta)$ is the overall denoising network with parameters Θ . The ADAM optimizer [99] was used to train the network with the setting of regular parameters $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$, and we set the learning rate 10^{-4} , the batch size 8, the number of training epochs 50. The proposed network is implemented on PyTorch and trained with 2 Nvidia 3090 Ti GPUs.

IV. EXPERIMENTS

A. Experimental Setup

During our experiments, we use the standard Vimeo dataset [35] to train the proposed deep unfolding network. The Vimeo dataset consists of 91701 sequences, which are classified into a training dataset and a testing dataset. To provide more comprehensive results, we introduce another high-definition test dataset in [67], i.e., DTMC-HD, in our comparisons. Specifically, the DTMC-HD testing dataset contains 7 different videos, and the videos of DTMC-HD have a resolution of 960×540 with 75 frames. Following the settings in [54], we consider the mixture of additive white Gaussian noise (AWGN) and impulse noise (IN), and there are two common types of IN, i.e., salt-and-pepper impulse noise (SPIN) and random-valued impulse noise (RVIN). The three types of mixed noise that we employ are AWGN + SPIN, AWGN + RVIN, and AWGN + SPIN + RVIN, respectively. In the following experimental settings, σ_n denotes the standard deviation of the Gaussian noise component, s and r are the SPIN and RVIN ratios, respectively. Note that all experiments are performed on RGB channels.

Taking into account a more realistic degraded scenario, we implement supplementary experiments to recover the clean

TABLE I
AVERAGE PSNR/SSIM RESULTS OF MIXED VIDEO NOISE REMOVAL (AWGN + SPIN, AWGN + RVIN AND AWGN + SPIN + RVIN)
IN DTMC-HD AND VIMEO DATASETS AT DIFFERENT NOISE LEVELS (BOLDFACE HIGHLIGHTS THE BEST)

Methods		V-BM4D* [30]	DeamNet [21]	FastDVDnet [40]	MAP-VDNet [67]	DU-MVDnet	DU-MVDnet-B
$\sigma = 10, s = 50\%$	DTMC-HD	31.42/0.8632	35.31/0.9231	35.62/0.9270	35.45/0.9254	35.73/0.9304	35.33/0.9253
	Vimeo	31.43/0.8804	38.17/0.9571	38.82/0.9625	38.52/0.9604	39.41/0.9673	38.79/0.9636
$\sigma = 20, s = 40\%$	DTMC-HD	31.04/0.8297	33.27/0.8895	33.28/0.8898	33.28/0.8909	34.18/0.9054	33.82/0.9008
	Vimeo	31.62/0.8461	35.68/0.9328	35.79/0.9335	35.73/0.9359	37.31/0.9520	36.80/0.9477
$\sigma = 30, s = 20\%$	DTMC-HD	31.44/0.8390	32.41/0.8728	32.87/0.8802	33.47/0.8883	33.60/0.8945	33.08/0.8853
	Vimeo	32.55/0.8629	34.68/0.9213	34.68/0.9233	36.23/0.9410	36.56/0.9464	35.92/0.9385
$\sigma = 10, r = 20\%$	DTMC-HD	33.41/0.9010	36.17/0.9316	36.14/0.9329	36.95/0.9385	37.09/0.9415	36.42/0.9364
	Vimeo	34.05/0.9218	39.30/0.9644	38.44/0.9599	39.58/0.9668	40.98/0.9749	40.11/0.9708
$\sigma = 20, r = 30\%$	DTMC-HD	29.27/0.8223	32.82/0.8810	33.07/0.8875	34.17/0.9019	33.95/0.9026	33.63/0.8966
	Vimeo	28.85/0.8199	35.31/0.9297	35.82/0.9326	37.17/0.9506	37.18/0.9518	36.73/0.9475
$\sigma = 30, r = 40\%$	DTMC-HD	25.26/0.7350	30.33/0.8315	31.05/0.8429	31.80/0.8573	31.82/0.8648	31.47/0.8523
	Vimeo	23.95/0.7137	32.61/0.8932	33.57/0.9006	34.41/0.9224	34.77/0.9275	34.22/0.9207
$\sigma = 5, s = 50\%, r = 5\%$	DTMC-HD	30.40/0.8586	35.93/0.9372	36.18/0.9393	37.08/0.9473	36.99/0.9475	36.39/0.9416
	Vimeo	30.21/0.8829	38.97/0.9641	39.09/0.9664	40.51/0.9740	40.91/0.9759	40.23/0.9733
$\sigma = 10, s = 40\%, r = 10\%$	DTMC-HD	30.38/0.8372	34.68/0.9180	35.07/0.9210	35.21/0.9225	35.62/0.9287	35.26/0.9243
	Vimeo	30.20/0.8480	37.52/0.9531	38.26/0.9588	38.15/0.9582	39.13/0.9659	38.62/0.9628
$\sigma = 15, s = 30\%, r = 15\%$	DTMC-HD	29.88/0.8185	33.50/0.8974	32.95/0.8789	33.83/0.9017	34.73/0.9156	34.29/0.9091
	Vimeo	29.78/0.8221	36.03/0.9368	36.35/0.9403	36.50/0.9441	37.96/0.9580	37.42/0.9536

image from compressed videos with mixed noise. Specifically, we first generate noisy frames with the mixed noise type of AWGN + RVIN for the level of $\sigma = 10, r = 20\%$, and we further compress noisy frames using the x265 standard of FFmpeg¹ with a constant quantization parameter $QP = 32$.

To verify the generalization of the proposed method, we have extended the developed network to the blind video decaptioning task. We use the ECCV Chalearn 2018 LAP Video Decaptioning Challenge dataset² as a training and testing dataset. In particular, this video decaptioning dataset has 100 videos containing both encrusted subtitles and without subtitles, and each video consists of 125 frames with a resolution of 128×128 . The captions in this dataset have different colors, sizes, positions, and shadows. Following the settings in [100], we use 70K sample pairs for training and 5K frames for testing, and all video clips are converted to PNG images to facilitate training.

B. Comparison With Other State-of-the-Art Methods

We have compared the proposed DU-MVDnet with several current state-of-the-art image and video denoising methods, including the model-based video denoising method, i.e., V-BM4D [30], and deep learning-based methods, i.e., DeamNet [21], FastDVDnet [40] and MAP-VDNet [67]. Among competing methods, the DeamNet is a deep learning-based image denoising method, while other algorithms are video denoising methods. Note that the V-BM4D is developed to deal with Gaussian noise; we follow a similar setting in [54] and develop an enhancement version to remove mixed video noise, denoted as V-BM4D*. In particular, we apply an adaptive median filter (AMF) [74] to remove SPIN and an adaptive center-weighted median filter (ACWMF) [76] to remove RVIN before the V-BM4D denoising algorithm. For

TABLE II
AVERAGE PSNR/SSIM RESULTS OF MIXED AND COMPRESSED VIDEO NOISE REMOVAL IN VIMEO AND DTMC-HD TEST DATASETS WITH THE MIXED AND COMPRESSED NOISE LEVEL OF $\sigma = 10, r = 20\%$, $QP = 32$ (BOLDFACE HIGHLIGHTS THE BEST)

	FastDVDnet	MAP-VDNet	DU-MVDnet
Vimeo	37.16/0.9471	39.35/0.9658	39.56/0.9677
DTMC-HD	34.41/0.9089	35.94/0.9272	35.93/0.9294

fair comparisons, we used a 3-stage model of MAP-VDNet to be consistent with the proposed model. In addition, we also trained a blind model of the proposed network, denoted as DU-MVDnet-B. To adapt to the blind denoising problem, the specific values of the standard deviation of the Gaussian noise σ_n , SPIN ratio s and RVIN ratio r are uniformly sampled from the ranges of $[0, 30]$, $[0, 50\%]$ and $[0, 50\%]$, respectively. The experimental results on two datasets are demonstrated in Table I. From these results, it could be observed that the proposed DU-MVDnet could obtain the best denoising performance compared to other state-of-the-art image and video denoising algorithms, i.e., DeamNet, FastDVDnet and MAP-VDNet. Furthermore, the DU-MVDnet-specific denoising model has demonstrated better denoising performance compared to the DU-MVDnet-B blind model. From the visual comparisons in Figs. 4, 5, and 6, one can see that the proposed network could recover sharper edges and preserve more details of the image.

C. Mixed and Compressed Video Noise Removal

To further consider the influence of compressed noise, we conduct comparative experiments to enhance the compressed videos with mixed noise. In particular, we have compared the denoising performance of the proposed method with the state-of-the-art deep learning-based video denoising methods, e.g., FastDVDnet [40] and MAP-VDNet [67], and the

¹<http://ffmpeg.org/ffmpeg.html>

²<https://chalearnlap.cvc.uab.cat/dataset/31/description/#>

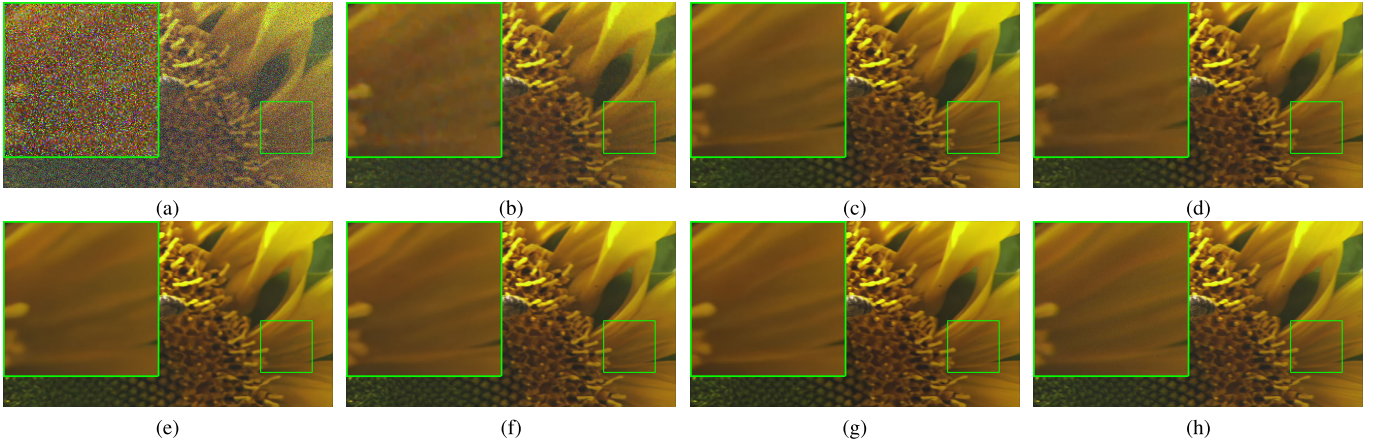


Fig. 4. Denoising results for a noisy frame of *Sunflower* video of the DTMC-HD test set with mixed noise level of $\sigma = 20, s = 40\%$. (a) Noisy frame; denoised by (b) V-BM4D* [30] (30.49 dB, 0.8658), (c) DeamNet [21] (35.24 dB, 0.9224), (d) FastDVDnet [40] (34.71 dB, 0.9207), (e) MAP-VDNet [67] (34.90 dB, 0.9224), (f) **DU-MVDnet (36.16 dB, 0.9323)** and (g) DU-MVDnet-B (34.08 dB, 0.9267); (h) Original frame.

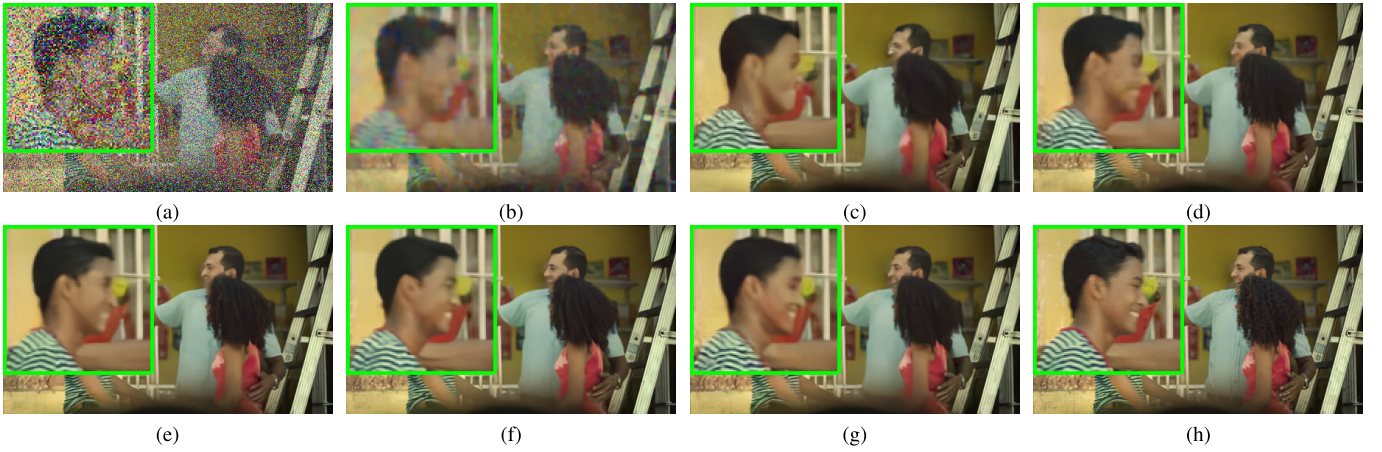


Fig. 5. Denoising results for a noisy frame from the Vimeo test set with a mixed noise level of $\sigma = 30, r = 40\%$. (a) Noisy frame; denoised by (b) V-BM4D* [30] (22.47 dB, 0.6847), (c) DeamNet [21] (28.37 dB, 0.8503), (d) FastDVDnet [40] (29.18 dB, 0.8729), (e) MAP-VDNet [67] (30.08 dB, 0.8922), (f) **DU-MVDnet (30.43 dB, 0.9015)** and (g) DU-MVDnet-B (29.97 dB, 0.8914); (h) Original frame.

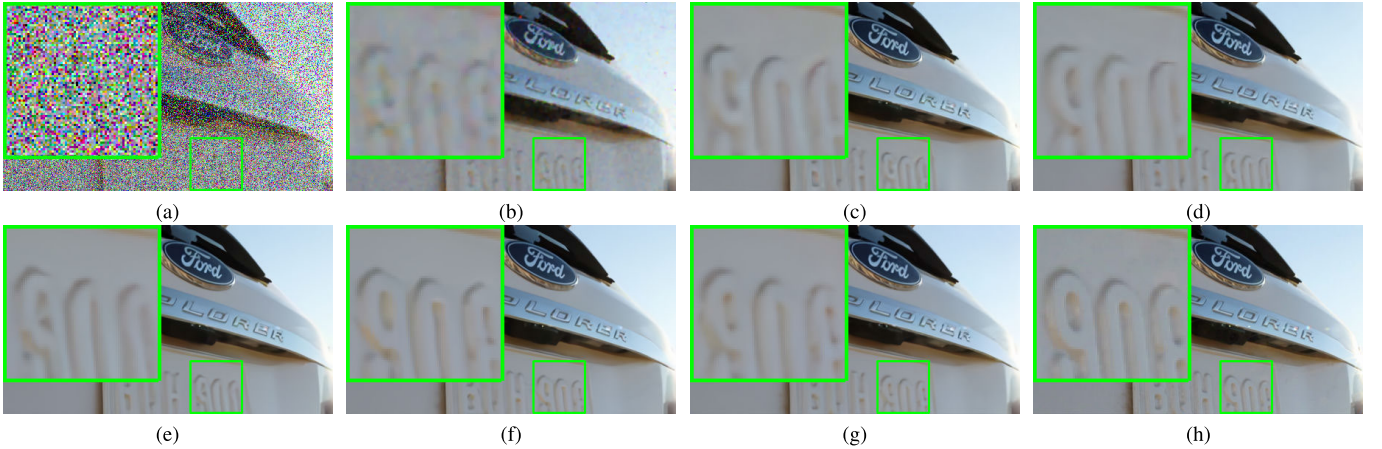


Fig. 6. Denoising results for a noisy frame from the Vimeo test set with the mixed noise level of $\sigma = 15, s = 30\%, r = 15\%$. (a) Noisy frame; denoised by (b) V-BM4D* [30] (30.67 dB, 0.8800), (c) DeamNet [21] (36.93 dB, 0.9600), (d) FastDVDnet [40] (36.26 dB, 0.9558), (e) MAP-VDNet [67] (37.37 dB, 0.9634), (f) **DU-MVDnet (38.21 dB, 0.9688)** and (g) DU-MVDnet-B (37.56 dB, 0.9652); (h) Original frame.

experimental results on Vimeo and DTMC-HD test datasets for the mixed and compressed noise level of $\sigma = 10, r = 20\%$, $QP = 32$ are demonstrated in Table II. From the comparative results, one can observe that the proposed method could obtain the most competitive denoising performance compared to FastDVDnet and MAP-VDNet. According to visual comparisons in Fig. 7, the developed DU-MVDnet could effectively remove

mixed and compressed noise and restore more details of the image, which has verified the effectiveness of the proposed method.

D. Video Decaptioning

To verify the generalization property of the proposed network, we have implemented extended experiments on

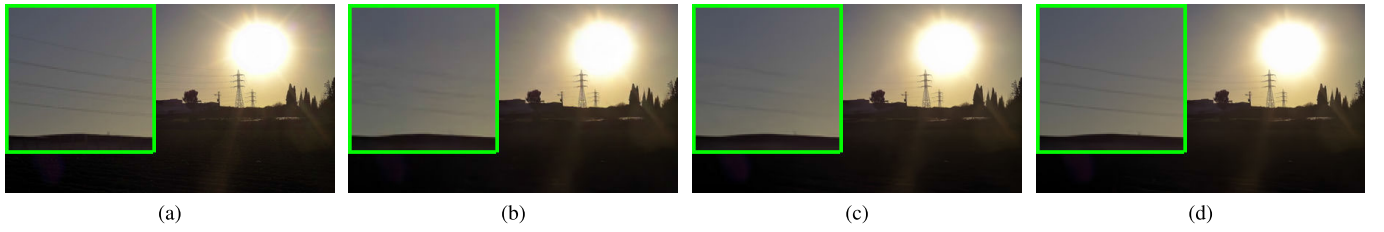


Fig. 7. Denoising results for a compressed and noisy frame from the Vimeo test set with the mixed and compressed noise level of $\sigma = 10, r = 20\%$, $QP = 32$. (a) Original frame; frame enhanced by (b) FastDVDnet [40] (39.88 dB, 0.9590), (c) MAP-VDNet [67] (40.98 dB, 0.9636) and (d) **DU-MVDnet** (41.57 dB, **0.9670**).

TABLE III

AVERAGE PSNR/DSSIM RESULTS OF VIDEO DECAPTIONING ON CHALEARN 2018 LAP INPAINTING TRACK2 VALIDATION SET (BOLDFACE HIGHLIGHTS THE BEST). [†] DENOTES THE RESULTS CITED FROM THEIR ORIGINAL PUBLICATIONS

Metric	PSNR	DSSIM	GFLOPs
BVDNet [†] [100]	34.71	0.0222	15.05
DVDNet [†] [101]	35.03	0.0317	25.60
DU-MVDnet	35.43	0.0310	12.07

TABLE IV

AVERAGE PSNR (dB) AND SSIM RESULTS OF COMPARATIVE EXPERIMENTS WITH AND WITHOUT $\tilde{s}_j$ PRIOR TERM FOR THE MIXED NOISE LEVEL $\sigma = 5, s = 50\%, r = 5\%$

Model	without $\tilde{s}_j$	with $\tilde{s}_j$
DTMC-HD	36.28/0.9421	36.99/0.9475
Vimeo	40.52/0.9746	40.91/0.9759

blind video decapting without input masks. Competing methods consist of the BVDNet [100] method and the DVDNet [101] method, which has achieved state-of-the-art blind video decapting performance. Note that the comparative results come from their publications, and the experimental results are demonstrated in Table III. From these comparative results, one can see that the proposed method could obtain competitive blind decapting performance even with the smallest computational burdens. Furthermore, from the comparative visualizations in Fig. 8, the proposed network could better remove captions and preserve image details, which has verified the great robustness and generalization of the proposed method.

E. Ablation Study

The effect of ℓ_1 -norm modeling of $\tilde{s}_j$: Based on the sparse characteristics of impulse noise, we employ a ℓ_1 -norm regularization for $\tilde{s}_j$ in Eq. (5). To prove the effectiveness of ℓ_1 -norm modeling of $\tilde{s}_j$, we implemented comparative experiments with and without the prior term $\tilde{s}_j$. Specifically, the model without the $\tilde{s}_j$ prior term is consistent with the i.i.d. MAP estimator in [67], and the main difference relies on not conducting the soft-thresholding computation in Fig. 1. For fair comparisons, we keep other modules (the progressive fusion and denoiser) the same as the proposed DU-MVDnet except for the weighted fusion operation. The experimental results have been summarized in Table IV, one could see that the proposed network could exactly prompt the denoising performance to varying degrees through the introduction of $\tilde{s}_j$ prior term.

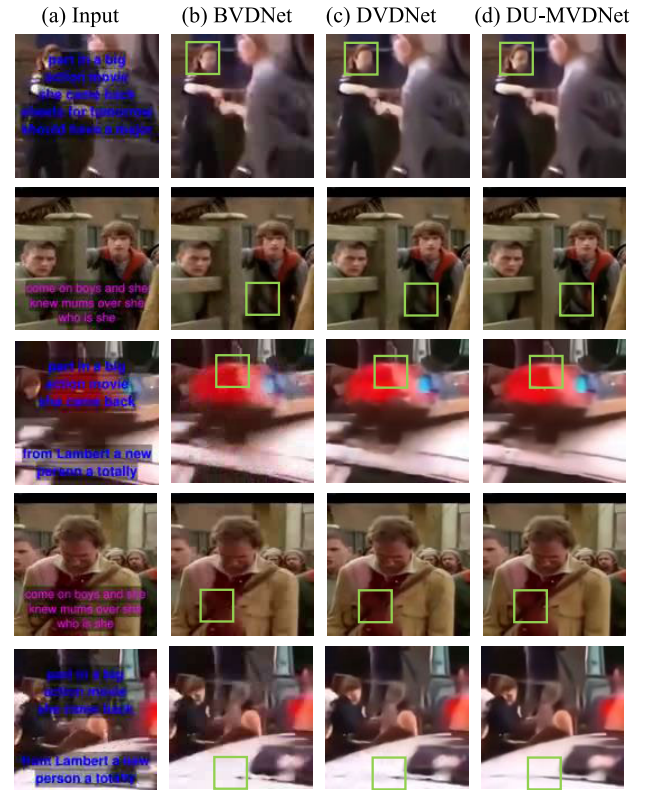


Fig. 8. Visual comparison of decapting frames recovered by: (b) BVDNet [100], (c) DVDNet [101] and (d) DU-MVDnet.

TABLE V

COMPARISON OF THE AVERAGE RESULTS OF PSNR (dB) AND SSIM WITH DIFFERENT INITIALIZATIONS OF $\mathbf{x}^{(0)}$ FOR THE MIXED NOISE LEVEL $\sigma = 15, s = 30\%, r = 15\%$

Initialization of $\mathbf{x}^{(0)}$	$\mathbf{y}_0$	$\tilde{\mathbf{y}}'$
DTMC-HD	34.43/0.9120	34.73/0.9156
Vimeo	37.81/0.9563	37.96/0.9580

The effect of the initialization of $\mathbf{x}^{(0)}$: To determine the appropriate initialization of $\mathbf{x}^{(0)}$, we conduct experiments using the reference noisy frame $\mathbf{y}_0$ and the aggregate frame $\tilde{\mathbf{y}}'$ to initialize $\mathbf{x}^{(0)}$, respectively, the comparative results for the mixed noise level $\sigma = 15, s = 30\%, r = 15\%$ are demonstrated in Table V. In principle, the initialization of $\mathbf{x}^{(0)}$ would influence the performance of the following soft-thresholding and denoising modules. According to the experimental results, the model using $\tilde{\mathbf{y}}'$ to initialize $\mathbf{x}^{(0)}$ could obtain greater denoising performance, which has verified our assumption.

TABLE VI
AVERAGE PSNR (dB) AND SSIM RESULTS OF VARIANTS OF THE
PROPOSED METHOD ON THE VIMEO TEST DATASET FOR
MIXED NOISE LEVEL $\sigma = 15$, $s = 30\%$, $r = 15\%$

Multi-stage progressive fusion module	Adaptive filters	PSNR	SSIM
✗	✓	36.40	0.9451
✓	✗	37.43	0.9462
✓	✓	37.96	0.9580

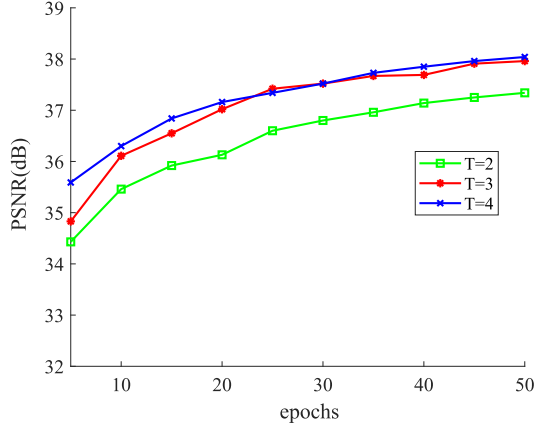


Fig. 9. Performance curves as a function of epochs in different stages of the Vimeo test dataset for the mixed noise level $\sigma = 15$, $s = 30\%$, $r = 15\%$.

The effects of main components: To verify the effectiveness of the proposed components (the multistage progressive fusion module and adaptive filters), we implement comparative experiments with and without these mechanisms; the results for the mixed noise level $\sigma = 15$, $s = 30\%$, $r = 15\%$ are demonstrated in Table VI. To make fair comparisons, for the model without a fusion module, we use a classic optical flow estimation network, i.e., Spynet [93], to replace the multistage progressive fusion, and for the model without adaptive filters, we exploit a common concatenation operation. From the comparisons in Table VI, it could be observed that the use of the progressive fusion module and adaptive filters could introduce performance improvements to different degrees, i.e., 1.56 dB and 0.53 dB, respectively, demonstrating the effectiveness of the proposed components.

The effect of the number of stages: To demonstrate the relationship between denoising performance and the number of iterations/stages T , we plot the PSNR curves of different stages to describe the change in denoising performance along with the increase in epochs. As shown in Fig. 9, one can see that the model with more stages could boost the denoising performance in general. However, more iterative computations would lead to the rapid growth of computational complexity, and denoising performance tends to be saturated when $T \geq 3$. To make a trade-off between denoising performance and computational burdens, we set the number of stages $T = 3$ in our implementation.

The effect of the number of input frames: In this paper, we take a fixed number of adjacent noisy frames as input to recover the reference frame. To present the influence of the number of input frames on the denoising performance, we implement comparative experiments in which the number

TABLE VII
COMPARISON OF THE AVERAGE PSNR (dB) AND SSIM RESULTS
WITH DIFFERENT NUMBERS OF INPUT FRAMES FOR THE
MIXED NOISE LEVEL $\sigma = 15$, $s = 30\%$, $r = 15\%$

Dataset	$J = 1$	$J = 2$	$J = 3$
DTMC-HD	34.24/0.9088	34.47/0.9120	34.73/0.9156
Vimeo	37.27/0.9522	37.57/0.9550	37.96/0.9580

TABLE VIII
COMPARISON OF GFLOPs AND THE NUMBER OF PARAMETERS

Method	DeamNet	FastDVDNet	MAP-VDNet	DU-MVDnet
GFLOPs	9.14	2.60	16.12	12.07
No. of parameters (M)	2.23	2.48	2.31	3.20

of input frames is 3, 5, 7, respectively, corresponding to the situations of $J = 1$, $J = 2$, $J = 3$. From the results in Table VII, we can see that the performance improves as the number of input frames increases. The increase in the number of input frames would lead to the rapid growth of computational complexity; To strike a good trade-off between performance and computation, we set $J = 3$ in the following experiments.

F. Computational Complexity Analysis

We have compared the computational complexity (e.g., GFLOPs and number of network parameters) between DeamNet, FastDVDNet, MAP-VDNet and the proposed DU-MVDnet, and the results are demonstrated in Table VIII. Note that the comparative experiments are conducted on a Nvidia Titan XP GPU. From these comparisons, it could be observed that the computational cost (GFLOPs) of FastDVDNet is the smallest, yet the trade-off is made between the denoising performance and the computational complexity. Furthermore, the number of parameters of MAP-VDNet is less than that of DU-MVDnet because the same network parameters are shared between different stages. Compared to MAP-VDNet, the proposed DU-MVDnet possesses fewer computation burdens and obtains greater denoising performance.

V. CONCLUSION

In this paper, we propose a model-guided deep network to remove mixed video noise. To the best of our knowledge, this is the first work to address the mixed video noise removal problem through the association of MAP estimation and DCNNs. Specifically, we first develop a novel objective function towards the problem of mixed video noise removal, and derive an iterative optimization algorithm to recover the clean frame. To benefit from the end-to-end training, we introduce a deep denoiser prior and devise the overall network structure guided by the developed iterative algorithm. Moreover, instead of using time-consuming motion estimation and motion compensation operations, we propose an efficient multistage progressive fusion strategy, namely, we classify input noisy frames into several groups with different motion rates and progressively aggregate these frames into one frame. In particular, we develop an adaptive multi-frame aggregation

operation for intra-group and inter-group fusions; the temporal feature maps could be adaptively aggregated through weighting filters predicted by DCNN. Extensive experimental results of different types of mixed noise on different testing datasets have verified the effectiveness and robustness of the proposed method. In addition, video decaptioning experiments have demonstrated the great generalization performance of the proposed DU-MVDnet.

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